

Introduction to Proximal Policy Optimization

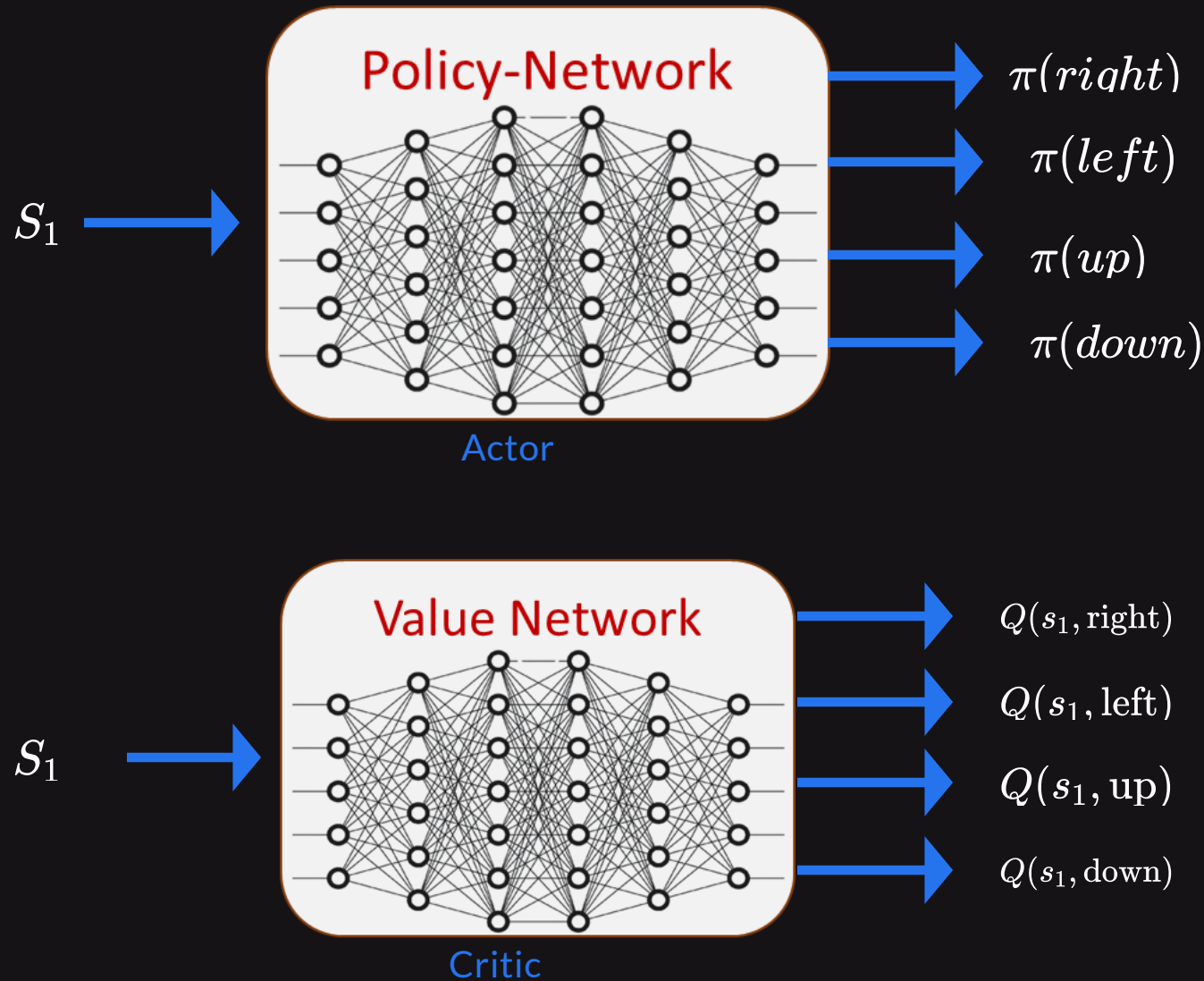
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Proximal Policy Optimization (PPO)



- We want to improve training stability by avoiding having too large policy updates
- PPO's Clipped Objective function:

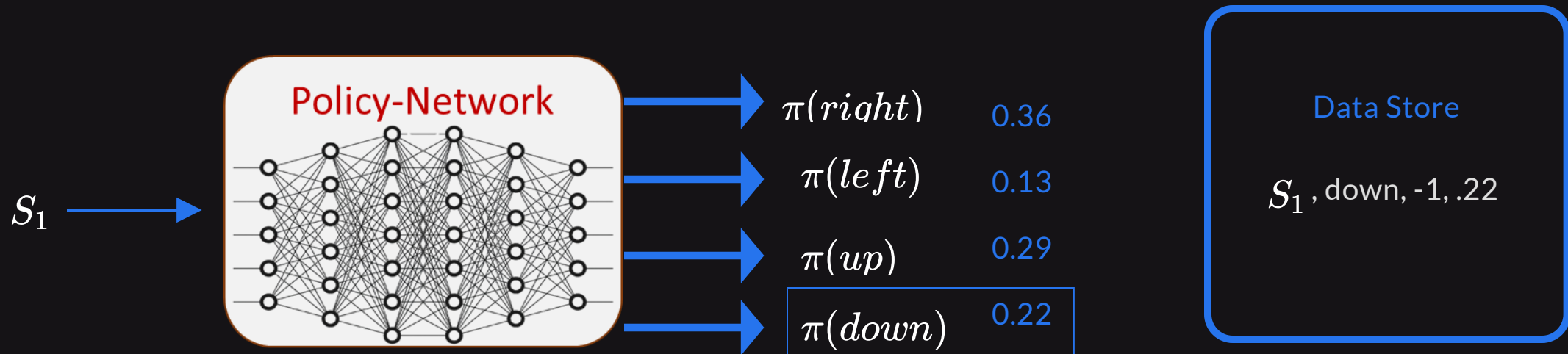
$$L^{CLIP} = \mathbb{E}_t [\min(r_t(\theta) A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) A_t)]$$

$$r_t(\theta) = \frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{old}}(a_t | s_t)}$$

Ratio Function

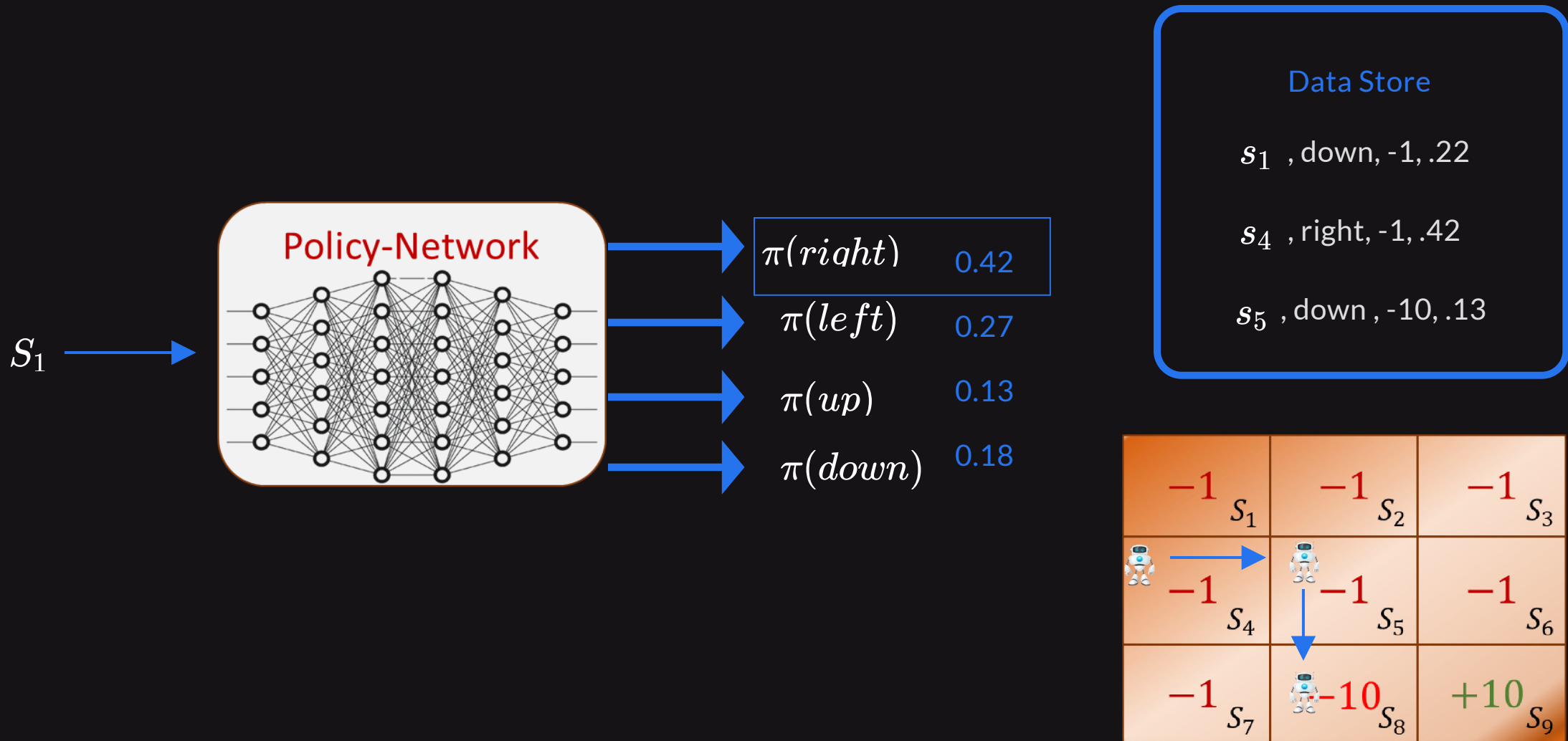
Advantage if $A_t > 0$, this action is better than the other action possible at that state.

Proximal Policy Optimization (PPO)

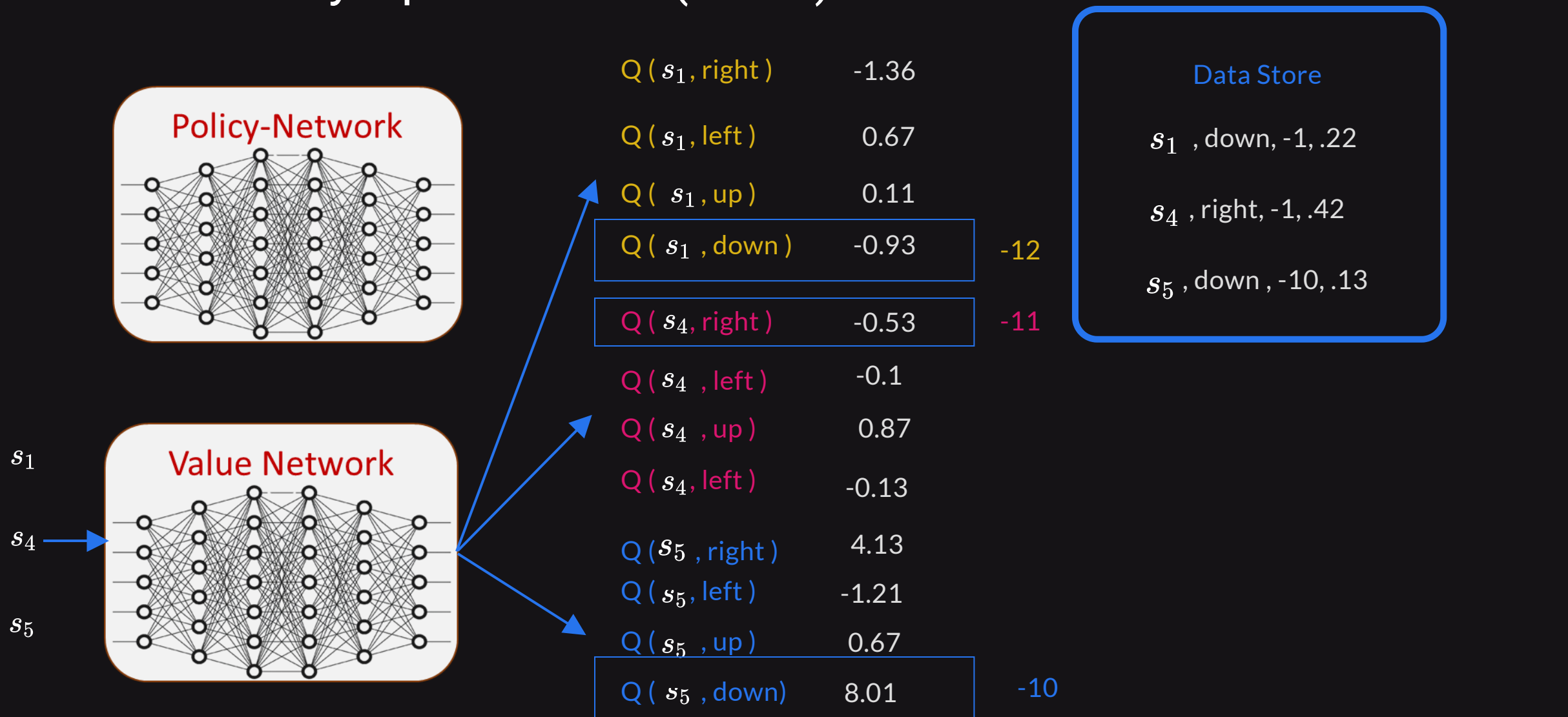


 -1 s_1	-1 s_2	-1 s_3
 -1 s_4	-1 s_5	-1 s_6
-1 s_7	-10 s_8	$+10$ s_9

Proximal Policy Optimization (PPO)



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$Q(s_1, \text{right})$	-1.36	
$Q(s_1, \text{left})$	0.67	
$Q(s_1, \text{up})$	0.11	
$Q(s_1, \text{down})$	-0.93	-12
$Q(s_4, \text{right})$	-0.53	-11
$Q(s_4, \text{left})$	-0.1	
$Q(s_4, \text{up})$	0.87	
$Q(s_4, \text{down})$	-0.13	
$Q(s_5, \text{right})$	4.13	
$Q(s_5, \text{left})$	-1.21	
$Q(s_5, \text{up})$	0.67	
$Q(s_5, \text{down})$	8.01	-10

$$\text{Advantage} = -0.93 - (-12) = 11.07$$

$$\text{Advantage} = -0.53 - (-11) = 10.47$$

$$\text{Advantage} = 8.01 - (-10) = 18.01$$

$$\text{Loss} = \frac{11.07^2 + 10.47^2 + 18.01^2}{3} = 185.51$$

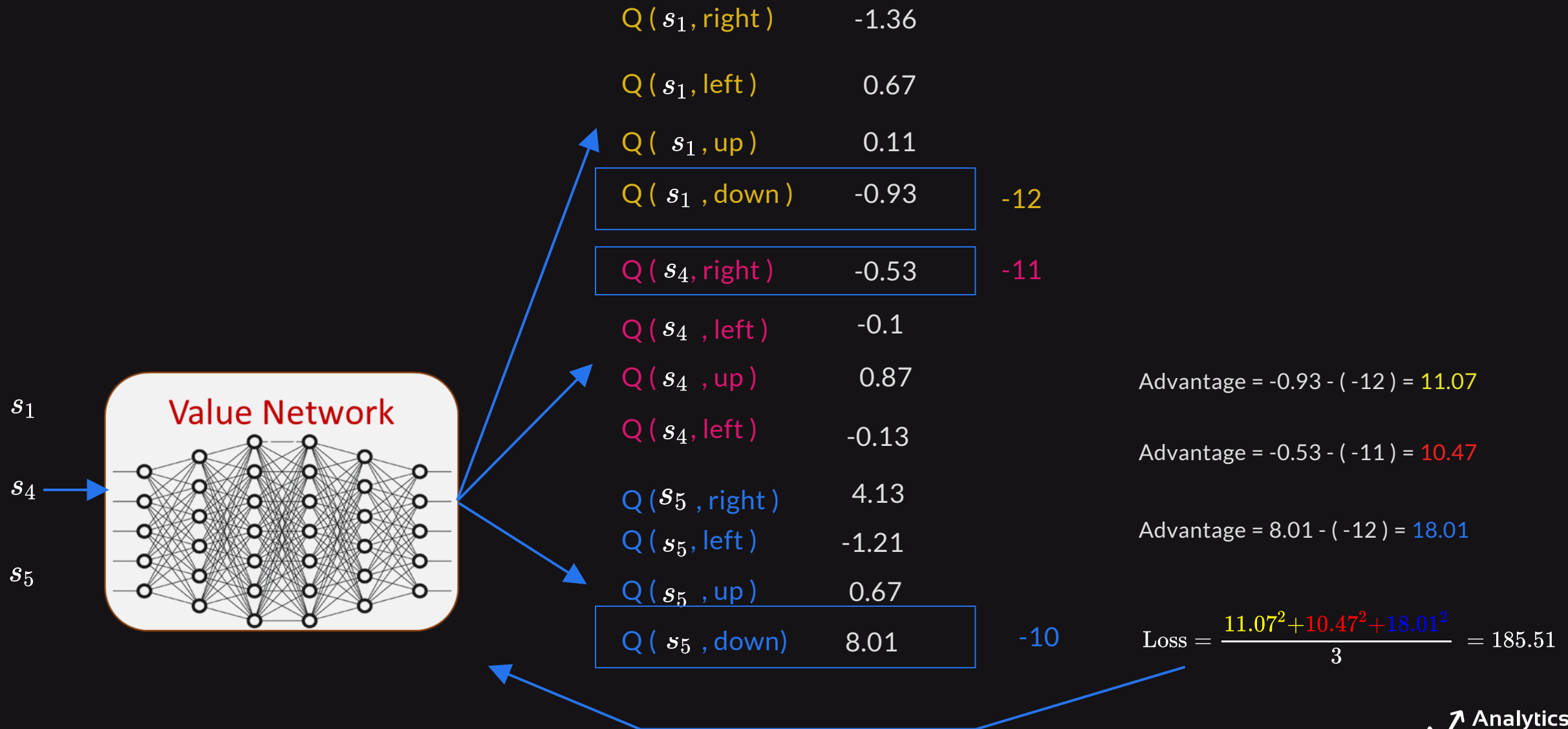
Data Store

s_1 , down, -1, .22

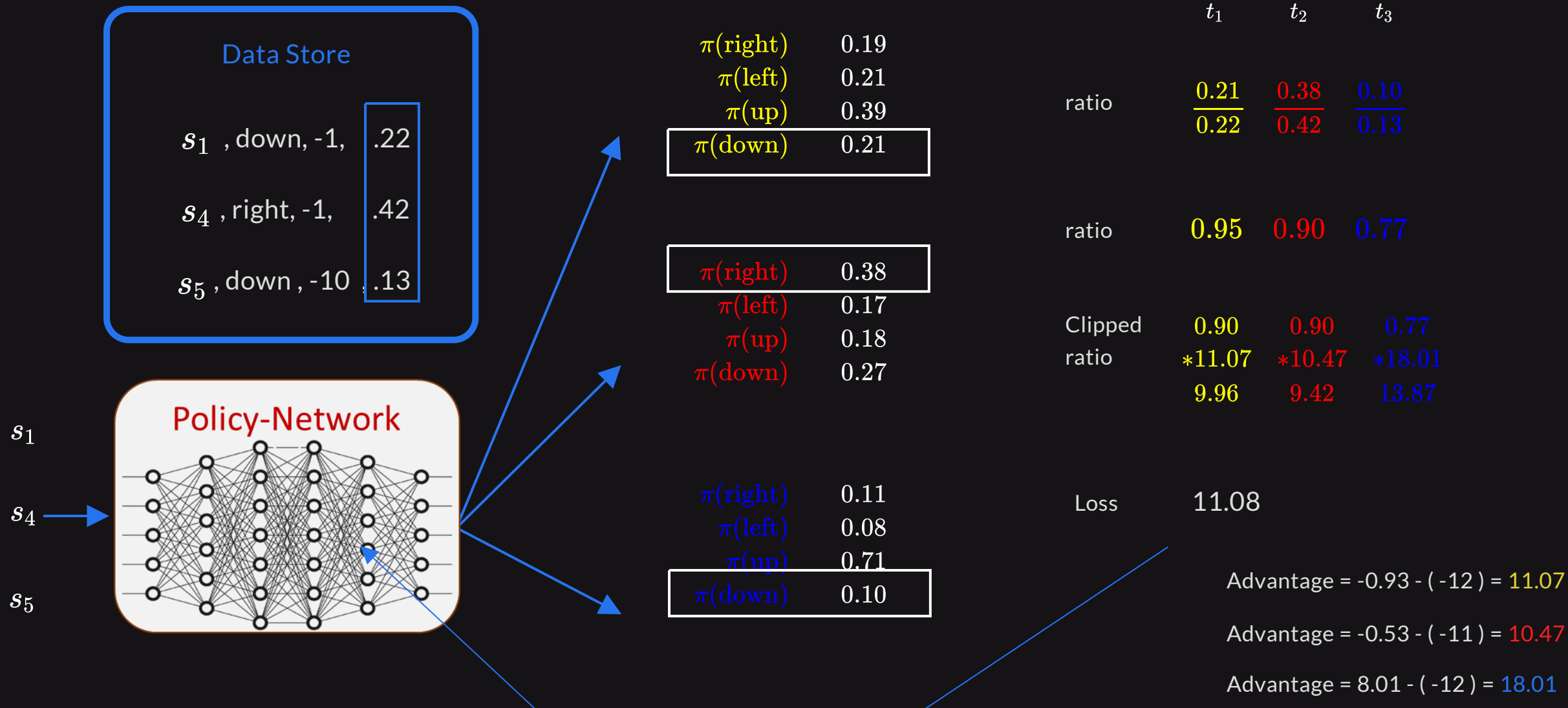
s_4 , right, -1, .42

s_5 , down , -10, .13

Proximal Policy Optimization (PPO)



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Thank You
